Machine Learning: k-Means and PCA for Regime Detection

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1. Abstract

In this project, we apply k-means clustering and PCA to solve the problem of market regime detection in multi-dimensional financial time-series. We compare these techniques against other regime detection methods: the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method. We evaluate the quality of each detection method using a trading strategy that goes long when the label indicates a bullish regime and flat otherwise, comparing cumulative returns across methods and regions (America, Europe, Asia).

2. Introduction

Financial markets, especially the stock market, have complexities. Due to various factors, such as macroeconomic recession, debt crisis, and policy changes, the market will show different regimes, such as bull and bear markets. Changes in market conditions often lead to fluctuations in stock prices. In different market regimes, investors need to use different trading strategies to increase returns and reduce losses. Therefore, detecting market regimes accurately plays an important role in making investment decisions and controlling trading risks. Traditional means of regime classification use a rules-based definition, such as the peak-to-trough regime labelling method. In contrast, unsupervised learning models identify regimes endogenously from the data, allowing for a more flexible and data-driven characterisation of market states.

K-means clustering and PCA are combined to identify market patterns in multidimensional financial time series. The trading strategy used to go long when the label shows bull and keep the position closed otherwise. Cumulative return, Annualized return, Annualized volatility and Sharpe ratio are used to measure the performance of trading strategies based on the results of market regime detection. The performance of the PCA k-means, the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method and long only strategy are compared in this project. In particular we fit our models on the period of 2010 to 2020 and we test the data on the period 2020 to 2024. We then compare the cumulative returns of the different strategies on the test set.

3. Data Processing and Feature Engineering

We track 9 major stock indices across 3 regions to develop our analysis from the 1st of January 2010 to 1st of January 2026.

Table 1: Index universe by region.

Region	Index	Ticker
America	S&P 500 / Dow Jones / Nasdaq	`^GSPC`, `^DJI`, `^IXIC`
Europe	FTSE 100 / CAC 40 / DAX	`^FTSE`, `^FCHI`, `^GDAXI`
Asia	KOSPI / Nikkei 225 / Hang Seng	`^KS11`, `^N225`, `^HSI`

3.1. Raw Data Structure

Each index is downloaded from Yahoo Finance with daily OHLCV columns. After merging, the full DataFrame uses a MultiIndex on columns of the form `(index_name, OHLCV_field)`.

3.2. Feature Engineering

From the closing prices we compute the following features for each index:

Log-returns. Over multiple horizons (1, 5, and 21 trading days):

$$r_{t,p} = \ln\left(\frac{P_t}{P_{t-p}}\right). \tag{1}$$

Rolling annualised volatility. With windows of 21 and 63 days:

$$\sigma_{t,w} = \mathrm{std}(r_{t-w:t}) \times \sqrt{252}\,. \tag{2}$$

Rolling correlation. For each pair of indices X and Y at time t with window w:

$$\rho_{X,Y,t} = \frac{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)(r_{Y,t-i} - \bar{r}_Y)}{\sqrt{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)^2 \ \sum_{i=0}^{w-1}(r_{Y,t-i} - \bar{r}_Y)^2}} \tag{3}$$

Momentum Z-score. Measures mean-reversion by normalising distance to the moving average:

$$Z_{t,w} = \frac{P_t - \mathrm{MA}_{t,w}}{\sigma_{t,w}} \tag{4}$$

RSI. The Relative Strength Index captures overbought and oversold conditions:

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \frac{\text{Average Gain}_t}{\text{Average Loss}_t}} \tag{5}$$

where Average Gain_t and Average Loss_t are computed over 14 days.

Building the feature set. We merge all features (log-returns, rolling volatility, correlation, z-score, RSI) into a single DataFrame. Per index: 3 log-returns + 2 volatilities + 1 z-score + 1 RSI = 7 features; across 3 indices this gives 21 features, plus 3 pairwise correlations, yielding a raw feature matrix of 24 columns. The number of rows equals the trading days minus a 63-day warm-up period. We standardise features and apply PCA retaining 95% of variance.

Table 2: Feature set dimensions after PCA (95% variance retained).

Region	Raw dimensions (Train + Test)	After PCA (Train + Test Data)
America	$(3459, 24)$	$(3459, 8)$
Europe	$(3527, 24)$	$(3527, 7)$
Asia	$(3564, 24)$	$(3564, 11)$

3.3. Pipeline Summary

Yahoo Finance API → Raw OHLCV → Merged DataFrame → Close prices

↓

Feature Engineering (returns, volatility, correlation, z-score, RSI)

↓

Standardisation → **PCA** → **K-Means Clustering**

Figure 1: Data pipeline from raw download to regime labels.

4. Mathematical Foundations

4.1. K-Means Clustering

K-means partitions data into K clusters by minimising the within-cluster sum of squares (WCSS):

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{6}$$

where C_k is the set of points in cluster k and μ_k is its centroid. In high-dimensional settings, the curse of dimensionality can limit performance; we therefore reduce dimensionality via PCA before clustering.

4.2. Principal Component Analysis (PCA)

PCA finds directions of maximum variance via the singular value decomposition of the centred data matrix:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{7}$$

where $\mathbf{U}$ contains left singular vectors, $\mathbf{\Sigma}$ is a diagonal matrix of singular values, and $\mathbf{V}$ contains the principal components. We retain the top d^* components capturing $\geq 95\%$ of variance.

4.3. Alternative Regime Detection Methods

To provide a broader comparison, we also evaluate several benchmark regime-detection methods with different modelling assumptions.

4.3.1 Gaussian-Mixture Hidden Markov Model (GMM-HMM)

The Hidden Markov Model assumes that the market evolves through a sequence of unobserved states, with state transitions governed by a first-order Markov chain:

$$P(z_t = j \mid z_{t-1} = i) = A_{ij} \tag{8}$$

where A_{ij} denotes the probability of moving from hidden state i to hidden state j. Conditional on the hidden state, the observed feature vector x_t is modelled as a Gaussian mixture:

$$p(x_t \mid z_t = k) = \sum_{m=1}^{M} \pi_{k,m} \mathcal{N}(x_t \mid \mu_{k,m}, \Sigma_{k,m}) \tag{9}$$

where $\pi_{k,m}$ are mixture weights, and $\mu_{k,m}, \Sigma_{k,m}$ are the corresponding mean vectors and covariance matrices. This allows the model to capture both temporal persistence in market regimes and more flexible within-state feature distributions.

4.3.2 Sliced-Wasserstein K-Means

Unlike standard K-means, sliced-Wasserstein K-means compares rolling return windows through their empirical distributions rather than Euclidean distances between daily feature vectors. Given two distributions μ and ν, the sliced-Wasserstein distance is defined as

$$SW_p(\mu,\nu) = \left(\int_{\mathbb{S}^{d-1}} W_p^p(P_\theta \# \mu,\ P_\theta \# \nu)\, d\sigma(\theta) \right)^{1/p} \tag{10}$$

where P_θ denotes projection onto direction θ, $P_\theta \# \mu$ is the projected distribution, and W_p is the one-dimensional Wasserstein distance. In practice, this is approximated using a finite number of random projections. This method is therefore more sensitive to changes in distributional shape and tail-risk structure across rolling windows.

4.3.3 Peak-to-Trough and Long-Only Benchmarks

As a simple rule-based benchmark, peak-to-trough labelling identifies crisis periods using drawdowns from the running peak of the cumulative return series:

$$DD_t = \frac{C_t - \max_{s \le t} C_s}{\max_{s \le t} C_s} \tag{11}$$

where C_t denotes the cumulative portfolio value at time t. A period is labelled as a crisis regime whenever DD_t falls below a chosen threshold. Finally, a long-only strategy is included as a passive benchmark that remains fully invested throughout the sample period.

5. Proposed Algorithm

Here we present a detailed algorithmic outline of the PCA K-Means regime detection pipeline, summarising the key steps from raw data processing to regime labelling and ordering.

Algorithm 1 K-Means Regime Detection Pipeline

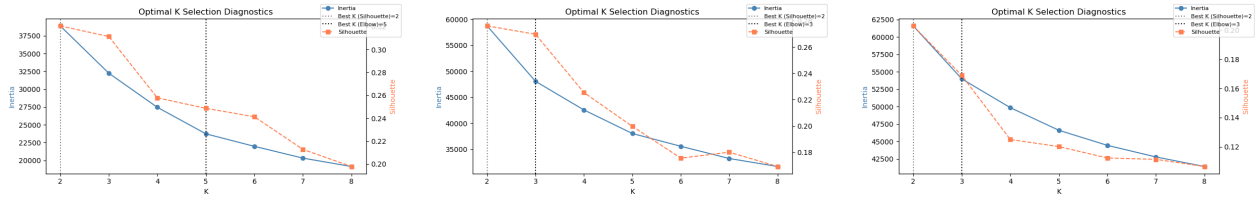
Require: Closing prices $\mathbf{P} \in \mathbb{R}^{T \times N}$, variance threshold α, range $[K_{\min}, K_{\max}]$
Ensure: Regime labels $\ell_t \in \{0, \ldots, K^* - 1\}$, ordered by mean return

— **Stage 1: Feature Engineering** —
for each index $n = 1, \ldots, N$ **do**
  Compute log-returns $r_{t,p}^{(n)}$ for $p \in \{1, 5, 21\}$
  Compute rolling volatility $\sigma_{t,w}^{(n)}$ for $w \in \{21, 63\}$
  Compute momentum z-score $Z_t^{(n)}$ and $\mathrm{RSI}_t^{(n)}$
end for
for each pair (i, j) with $i < j$ **do**
  Compute $\rho_t^{(i,j)} \leftarrow \mathrm{RollingCorr}_{63}(r^{(i)}, r^{(j)})$
end for
Concatenate into $\mathbf{F} \in \mathbb{R}^{T' \times D}$; drop warm-up rows
— **Stage 2: Standardisation & PCA** —
Z-score normalise each feature; compute SVD of $\tilde{\mathbf{F}}$
Select $d^* = \min\{d : \sum_{i=1}^{d} \sigma_i^2 / \sum_{i=1}^{D} \sigma_i^2 \ge \alpha\}$; project $\mathbf{X} \leftarrow \tilde{\mathbf{F}}\, \mathbf{V}_{:,1:d^*}$
— **Stage 3: Optimal K & Clustering** —
for $K = K_{\min}, \ldots, K_{\max}$ **do**
  Run K-Means; record inertia $\mathcal{I}(K)$ and silhouette $\mathcal{S}(K)$
end for
$K^* \leftarrow \arg\max_K \mathcal{S}(K)$; run final K-Means with K^*
— **Stage 4: Regime Ordering** —
Compute mean basket return $\bar{r}_k$ per cluster; relabel 0 = bearish, $K^* - 1$ = bullish
return $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{T'})$

6. Results

6.1. Optimal K Selection

We applied k-means for $K \in \{2, \ldots, 10\}$ on the PCA-reduced feature space and computed inertia and silhouette scores. The silhouette score is maximised at $K = 2$ across all regions, suggesting two dominant regimes over 2010–2020. This result suggests that, given the sample period and feature design adopted in this study, market states are primarily characterised by two dominant structures: a relatively weak bearish regime and a relatively strong bullish regime. Relative to larger values of K, the two-regime specification achieves better cluster separation while avoiding the introduction of excessive noise and potential overfitting. It also preserves the stability and interpretability of the detected regimes.

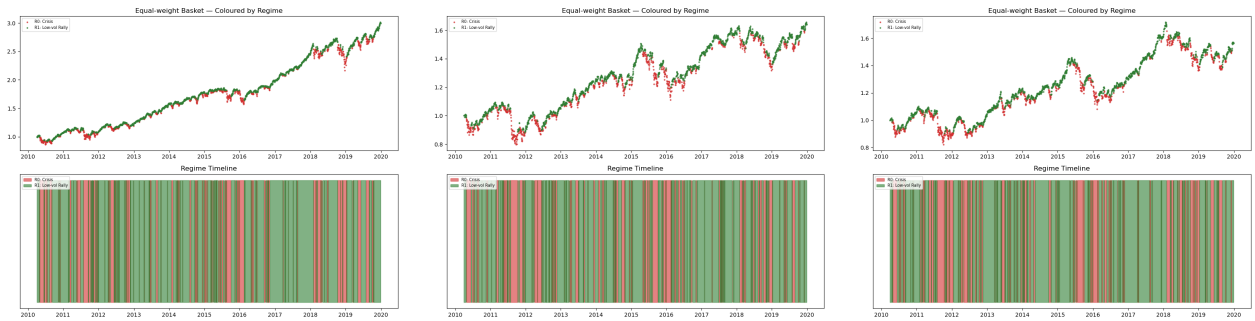


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 2: Inertia and silhouette scores for different K values across regional indices.

6.2. Regime Detection Results

Figure 3 shows the detected regimes overlaid on an equally weighted basket of indices for each region during the period 2010–2024. In the figure, the red segments represent the crisis regime identified by the model, while the green segments indicate more favourable regimes. These periods broadly align with major historical episodes of market stress, including the 2011 European debt crisis, and the 2020 COVID-19 pandemic.

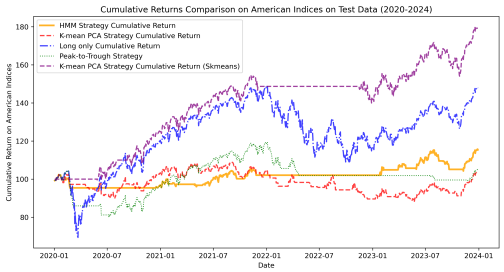


(a) American indices. **(b)** European indices. **(c)** Asian indices.

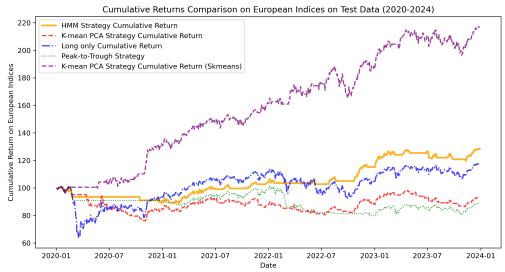
Figure 3: Regime detection on equally weighted baskets for each region.

6.3. Trading Strategy Comparison

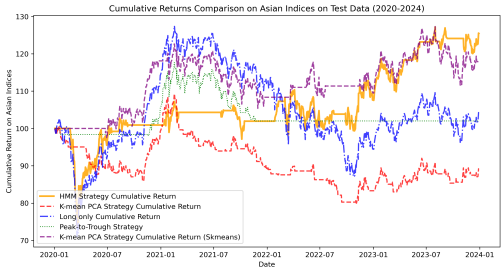
To evaluate regime quality, we implement a long/flat strategy: go long in bullish regimes, hold cash otherwise. We benchmark the PCA K-Means model against GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling (20% drawdown threshold), and a long-only baseline. We used the 90% variance threshold for PCA and we used Return periods of 1, 5, and 21 days, rolling volatilities with windows of 21 and 63 days, momentum z-score with a window of 21 days and RSI as features for the PCA K-Means model. We fitted and trained the model on the period of 2010 to 2024. We then tested the data on the period 2020 to 2024. On the test set the training strategies start with an inital capital of \$ 100 at the 1st of January 2020 and we calculate the cumulative returns until the 1st of January 2024. We shift the signal by one day to avoid look-ahead bias, and we do not apply transaction costs or slippage in this backtest. We plot in the following the performance of the PCA K-Means strategy in red against other strategies on the test period (2020–2024) in Figure 4.



(a) American indices.



(b) European indices.



(c) Asian indices.

Figure 4: Out-of-sample cumulative returns of trading strategies based on different regime detection methods.

The PCA K-Means strategy underperforms consistently more complex models such as the HMM, sW K-Means as well as the long-only regime detection models as shown in Table 3.

Table 3: Out-of-sample backtest (2020–2024) performance summary across regional indices.

Region	Metric	PCA K-Means	HMM	sWK-Means	Peak-to-Trough	Long Only
American	Cumulative Return (%)	103.65	115.39	169.39	104.76	147.80
	Annualised Return (%)	1	4	14	1	10
	Annualised Volatility (%)	11	6	18	12	23
	Sharpe Ratio	0.14	0.60	0.83	0.16	0.54
European	Cumulative Return (%)	92.96	128.35	216.95	88.86	117.51
	Annualised Return (%)	−2	6	21	−3	4
	Annualised Volatility (%)	11	7	13	11	20
	Sharpe Ratio	−0.11	0.87	1.58	−0.22	0.30
Asian	Cumulative Return (%)	89.13	125.46	117.90	102.01	104.19
	Annualised Return (%)	−3	6	4	0	1
	Annualised Volatility (%)	10	15	12	7	18
	Sharpe Ratio	−0.22	0.44	0.39	0.11	0.14

Looking at the performance metrics, the sWK-Means strategy achieves the highest cumulative returns across regions except the Asian region (169.39%, 216.95%, 117.90% for America, Europe, and Asia respectively). The HMM strategy delivers consistently strong risk adjusted performance in our framework. However, our main PCA K-Means strategy underperforms the long-only benchmark in America and Asia, and it has a negative Sharpe ratio in Europe and Asia, indicating that its regime signals are not effectively capturing profitable trading opportunities in these regions. The peak-to-trough method is worse than the PCA K-Means methods only for the European market but still outperforms our model for the American and Asian markets.

We can therefore see that the PCA K-Means methods, while conceptually appealing and interpretable, do not perform as well as more complex methods that either capture temporal dependencies (HMM) or distributional changes (sWK-Means). This suggests that the regimes detected by PCA K-Means may be too noisy or not sufficiently aligned with market conditions that drive returns, while the more sophisticated methods are better able to extract meaningful regime signals from the data. Its underperformance against long-only benchmarks in the three regions, indicates this major flaw in its regime detection capacity.

Regional differences. The PCA K-Means strategy performs relatively better in the American market, achieving a positive Sharpe ratio. In contrast, it underperforms in Europe and Asia, with negative Sharpe ratios. This pattern may reflect differences in market structure and co-movement across regions. The American indices are more tightly correlated, which may lead to clearer regime separation in feature space, while the Asian indices are more heterogeneous, making regime detection more challenging.

6.4. Hyperparameter tuning

We perform hyperparameter tuning for the PCA K-Means model to identify the preferred specification across different regional markets. The tuning is carried out using a grid search over several key parameters, and the corresponding hyperparameter grid is shown in Table 4.

Table 4: Hyperparameter grid for PCA K-Means tuning.

Hyperparameter	Values
n_regimes	$\{2, 3, 4, 5\}$
`use_pca`	{`True`, `False`}
`pca_variance`	$\{0.90, 0.95\}$
`return_periods`	{[1,5,21], [1,5,21,63]}
`vol_windows`	{[21,63], [10,21,63]}

The hyperparameters are defined as follows.

- `n_regimes`: the number of market states the clustering model is allowed to identify.
- `use_pca`: whether PCA is applied before clustering.
- `pca_variance`: how much of the total variance is retained by PCA.
- `return_periods`: the time horizons used to construct return features.
- `vol_windows`: the time windows used to construct volatility features.

Each parameter combination is evaluated using both the silhouette score and the Sharpe ratio of the corresponding regime-based trading strategy. The silhouette score measures how clearly separated the detected clusters are, while the Sharpe ratio measures the risk-adjusted return of the resulting strategy. To balance clustering quality and trading usefulness, we defined a combined score as an equally weighted average of the silhouette score and a rescaled Sharpe ratio:

$$\text{Combined Score} = 0.5 \times \text{Silhouette} + 0.5 \times \frac{\text{Sharpe}}{3}. \tag{12}$$

Table 5 reflects the preferred configuration within the PCA K-Means framework, rather than the overall ranking of this method against all other models.

Table 5: Final PCA K-Means parameters by region

Region	n_regimes	use_pca	pca_variance	return_periods	vol_windows
America	2	True	0.9	[1,5,21]	[10,21,63]
Europe	4	True	0.9	[1,5,21]	[21,63]
Asia	3	True	0.9	[1,5,21]	[10,21,63]

The tuning results show several common patterns across the three regions. First, all regions prefer using PCA with 90% of the variance, suggesting that moderate dimensionality reduction helps remove redundant information and noise while preserving the main clustering structure. Second, all three regions select return periods of $[1, 5, 21]$, rather than including a 63-day return horizon implying that short- and medium-term return dynamics already contain most of the information for regime detection in our framework. The main regional differences appear in the number of regimes and the volatility windows. The American market prefers 2 regimes, consistent with a relatively clear

two-state structure such as bullish versus bearish conditions. The Asian market prefers 3 regimes, while the European market prefers 4, indicating that these markets require finer distinctions in regime structure. This suggesting that a simple binary bull/bear split may be too coarse to capture the full variation in market behaviour over the sample period. This also suggests that the simpler structure favoured by silhouette score alone does not necessarily match the specification that performs best under the combined score. Therefore, the $K = 2$ result in Section 6.1 is better interpreted as a benchmark for cluster structure, while Table 5 reflects the more suitable full parameter combination for each region under the combined evaluation criterion.

The volatility-window resultsalso differ across regions. Both the American and Asian markets prefer $[10, 21, 63]$, suggesting that shorter-term volatility information adds value for regime detection in these markets, and that regime changes may be more sensitive to short-run risk fluctuations. In contrast, the European market prefers $[21, 63]$, indicating that smoother medium-term volatility measures are more informative than short-term volatility for capturing changes. Overall, the PCA K-Means framework is fairly stable across regions, since the use of PCA, the PCA variance threshold, and the return-period setting are consistent across all three regions. However, the optimal number of regimes and volatility windows still differ clearly by region, suggesting that the specific feature design remains influenced by differences in regional market structure.

7. Limitations

Our analysis is limited in several key ways. Firstly, all model classification performance is assessed on the results from the trading strategy. Whilst this is a useful proxy, it assumes that true regime classification translates to superior trading performance. In future work, the model performance could be tested using alternative methods. For instance, testing the models' regime classifiers against traditional rules-based methodologies such as drawdown or peak-to-trough labelling, allowing classification accuracy to be tested more directly.

Moreover, all trading performance was measured in the absence of transaction costs. This is an unrealistic expectation, and may have overstated the performance of models which frequently switch regimes. For future work, modelling these costs, such as bid-ask spreads with the Roll Model, may provide a more realistic assessment of model performance.

Another limitation concerns the lack of depth for training and testing of the models. We tested on a four-year period from 2020 to 2024. Although this was a highly volatile and varied period (e.g. COVID and subsequent recovery), a four-year window for testing limits the reliability and robustness of results. In future, increasing the size of the testing and training window would offer greater insight into the performance of the models. In addition, evaluating performance in varying market conditions would give more understanding on how the models adapt to different enviroments.

8. Conclusion

We have presented a PCA-enhanced k-means clustering pipeline for market regime detection across American, European, and Asian equity indices. The method identifies two dominant regimes (bullish and bearish) that align relatively well with known periods of market stress. However, when used to drive a simple long/flat trading strategy in out-of-sample test data, the PCA K-Means approach underperforms compared to alternative methods (GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling) and the long-only benchmark across all three regions. These results suggest that combining dimensionality reduction with unsupervised clustering provides a robust framework for capturing market dynamics. Additionally, hyperparameter tuning results show that most of the model specifications are optimal in all three markets, with PCA mainly providing useful improvements by removing redundant variation. However, more aggressive dimensionality reduction can further reduce noise and preserve key structures in high-dimensional time series. The results also show that the market regimes shows obvious regional differences, and the current feature set can well reflect

the market regimes. One limitation of this study is that there is no limit on the number of trades, and regimes can change very quickly when the market is volatile. Additionally, frequent trading incurs a lot of costs, which can affect the performance of the trading strategy.

9. Team Work and Contributions

9.1. Author Contributions (CRediT)

Conceptualisation:

Methodology: Oscar Peyron

Implementation: Oscar Peyron, Daniel Greenstein

Data Processing & Feature Engineering: Oscar Peyron

Model Training: Siqi Zhang

Analysis: Oscar Peyron

Validation & Testing: Stefaan Fernando, Siqi Zhang

Visualisation: Stefaan Fernando

Debug: Siqi Zhang, Zhidian Jing

Writing – Original Draft: Oscar Peyron, Zhidian Jing, Siqi Zhang

Validation & Testing: Stefaan Fernando

Visualisation: Stefaan Fernando

Writing – Original Draft: Oscar Peyron, Daniel Greenstein, Zhidian Jing

Writing – Review & Editing: Daniel Greenstein, Zhidian Jing

Project Administration: Daniel Greenstein

9.2. Reflection on Team Work

For this project, our team focussed on evaluating machine learning techniques on regime detection, with a primary focus on K-means and PCA. We settled on this project after a group polling system where various ideas were suggested, before selecting the majority vote. This ensured that the motivation for the group work was not driven by a single individual.

We split the work into several sections: model selection and coding, backtesting, hyperparameter selection, project write up and overall review. Tasks were then distributed based on individual preferences and skillsets (eg. coding ability, machine-learning knowledge, write-up ability) to assign work in the most effective way and in areas where members were most comfortable.

We utilised Teams to discuss what work had been performed and what required future attention. Meetings were conducted frequently, to track progress and align on next steps. Individual roles were tweaked and refined as workload evolved. In addition, we set up a WhatsApp group for discussion of specifics and individual queries. This enabled quick clarification of issues without requiring full meetings. A GitHub repository was also set up to provide a centralised workspace. This ensured each individual could work on up-to-date versions of the material, and live changes could be observed.

The team worked effectively together, and the individual contributions were well divided. In addition, the democratic selection for the project ensured that motivation and participation were high throughout.

Occasionally, competing coursework deadlines created a slight workload imbalance in the team, although this did not signifcantly impact overall progress. For future, an explicit review of deadlines should be conducted at the start of the project, to aid planning and workload distribution more efficiently. That said, overall contribution from all members was fair and satisfactory.

One challenge we did encounter was the finding of unreasonably strong trading returns. Subsequent analysis found some of the code had incorporated a form of look-ahead bias. The models had been trained on future data, creating falsely optimistic results. This was immediately notified to all members, and the data pipeline was corrected. Ultimately, trading returns and performance metrics(eg. Sharpe) began to show more realistic outcomes.

Overall, the team are happy with the outcome of the project and satisfied with the work of each individual.

Individual summary:

- **Zhidian Jing:** *[roles and concrete outputs to be added]*
- **Siqi Zhang:** splitting the dataset into training and test sets, training and testing the PCA + K-means model and the sWkmeans model, fixing the look-ahead bias issue, and modifying the sWkmeans output to a daily basis. I also obtained the hyperparameter tuning results from the tuning code and wrote the corresponding analysis in the report. I wrote Section 4.3 of the report.
- **Stefaan Fernando:** Implementation of the hyperparameter tuning pipeline for the K-Means regime detection model. Wrote a grid search script that systematically tested 48 combinations of hyperparameters including number of regimes, PCA settings, return periods and volatility windows. Each combination was scored using a combined metric of silhouette score and Sharpe ratio.
- **Daniel Greenstein:** Implementation of the Hidden Markov Model with Gaussian mixtures. This involved model fitting, parameter specificaiton and methodology write up. Involved in the write up, particularly for limitations, reflections and review. Involved in project administration, organising meetings.
- **Oscar Peyron:** Implementation of the PCA K-Means pipeline for regime detection. This included data processing, feature engineering, model fitting, and backtesting of the trading strategy. Oscar performed preliminary hyperparameter tuning for the number of regimes. Wrote part of the original draft, particularly the methodology and results sections.

9.3. AI Statement

Generative AI tools (Claude, Gemini, ChatGPT) were used in two capacities during this project:

1. **Code development:** AI assistance was used to generate baseline code, including boilerplate data-loading routines and initial model templates. All AI-generated code was reviewed, tested, and adapted by team members before inclusion in the final codebase.
2. **Report writing:** AI tools were used to assist with drafting and editing sections of this report, including improving clarity, grammar, and structure. All AI-generated text was critically reviewed and revised by the authors to ensure accuracy and consistency with the project's findings.